Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Enabled Home Mobile Manipulation Technology for Robust Object Search and Rearrangement			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	*Robotics/ Automation Machinery*	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	The proposed technology aims to develop an **AI-driven home mobile manipulation system** that enables a robot to autonomously search for, recognize, and rearrange everyday household objects in domestic environments. The system leverages advanced artificial intelligence to interpret high-level task requests, perceive complex home environments, and generate robust navigation and manipulation behaviors. Key features:			

	• Vision-language-based perception and reasoning for object detection, identification, and semantic understanding of home environments • Learning-based navigation and object search that operates reliably under incomplete, cluttered, and dynamic household conditions • AI-enabled mobile manipulation and skill coordination for stable execution of simple rearrangement tasks
Rationale for Joint R&D Project	The proposed project requires close collaboration between a research institute and an industrial partner to successfully translate AI-driven mobile manipulation technologies into a practical home robotics system. A joint R&D framework is essential to integrate advanced AI algorithms with real robotic hardware and to validate the technology under realistic domestic conditions. Rationale for collaboration: • The research institute provides expertise in AI-based perception, navigation, and mobile manipulation algorithms • The Korean SME partner provides capabilities in robot hardware development, system integration, and real-world experimentation • Effective co-design of AI software and robotic hardware is necessary to meet cost, safety, and reliability constraints Expected benefits of joint R&D: • Accelerated transition from AI research to deployable robotic prototypes • Reduced technical risk through iterative hardware–software co-development and validation • Creation of a scalable foundation for commercialization of AI-enabled home service robots
Key Words	1. AI-Driven Mobile Manipulation 2. Vision-Language-Based Object Search 3. Robust Home Robot Navigation 4. Adaptive Skill Coordination 5. Home Service Robotics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop an AI-driven home mobile manipulation system that can reliably operate in real-world domestic environments by integrating perception, navigation, and manipulation through advanced artificial intelligence. The specific R&D objectives are: ● To develop AI-based navigation algorithms that enable highly reliable robot movement and task execution in complex home environments ● To develop AI-based object detection and recognition using vision-language models for robust object search under uncertainty ● To develop AI-enabled mobile manipulation and skill coordination for stable and safe execution of everyday object rearrangement tasks			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Navigation Performance (99%>):** Achieve a navigation success rate greater than 99% in cluttered and dynamic home environments, including narrow passages and frequent human presence. **Object Detection and Search Performance (80%):** Achieve an object detection and identification success rate greater than 80% for common household objects using AI-based vision-language perception under partial observability. **Mobile Manipulation Performance (80%):** Achieve a manipulation success rate of at least 80% for simple household rearrangement tasks, such as picking, placing, and organizing objects, while maintaining stable and safe operation.			

R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to laboratory environments and finally to real-world domestic settings.
	
	● Year 1: Simulation-Based Development and Core AI Technology Implementation
	  ○ Develop an integrated simulation environment for home mobile manipulation tasks
	  ○ Implement AI-based object detection and semantic navigation using vision-language models
	  ○ Implement AI-enabled mobile manipulation and adaptive skill coordination for basic object handling
	  ○ Validate core algorithms through simulation-based experiments and performance benchmarking
	● Year 2: Laboratory-Scale Integration and Human-Interaction-Oriented System Design
	  ○ Integrate navigation, perception, and manipulation modules on a mobile manipulation platform in a laboratory environment
	  ○ Develop app- and language-based user interfaces for intuitive human–robot interaction			

	○ Customize and optimize AI algorithms for real hardware constraints and sensor noise ○ Conduct repeated laboratory experiments to evaluate robustness, safety, and task success rates ● Year 3: Real-World Deployment and Technology Advancement ○ Deploy the integrated system in realistic home environments ○ Perform large-scale experiments to verify navigation, object search, and manipulation performance ○ Refine system reliability, stability, and usability based on experimental results ○ Assess commercialization potential and prepare the technology for industrial transfer

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of core AI algorithms and system-level intelligence for the proposed home mobile manipulation technology. The institute will lead the research on perception, decision-making, and control methodologies required to achieve reliable robot operation in domestic environments. The specific roles and responsibilities of the research institute include: ● Development of AI-based navigation, object search, and mobile manipulation algorithms, including vision-language-based perception and adaptive skill coordination ● Design of the overall system architecture and integration framework for AI-driven mobile manipulation ● Simulation-based development, algorithm validation, and performance analysis of the proposed technologies Technical guidance and collaboration with the industrial partner to support system integration and experimental evaluation
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the robotic hardware and system-level implementation required for the proposed home mobile manipulation technology. The SME will play a key role in translating AI algorithms into a practical and deployable robotic system. The specific roles and responsibilities of the Korean SME partner include: ● Design and development of mobile manipulation hardware, including the mobile base, robotic arm, sensors, and system components ● System integration of AI algorithms with robotic hardware, sensors, and embedded software platforms ● Execution of laboratory-scale and real-world experiments, including data collection, testing, and performance evaluation ● Contribution to technology refinement, reliability improvement, and commercialization planning, leveraging industrial expertise and manufacturing capabilities

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver an AI-driven home mobile manipulation prototype capable of reliable object search and rearrangement in real-world domestic environments. The project will advance the state of the art in AI-based navigation, vision-language perception, and mobile manipulation, achieving measurable performance improvements in reliability and robustness. From an industrial perspective, this project will provide the Korean SME partner with core AI-enabled robotic technologies and system integration know-how applicable to next-generation home service robots. The outcomes will support **technology commercialization and product development** to enable cost-effective and scalable robotic solutions for domestic applications. More broadly, the project is expected to strengthen the competitiveness of Korean robotics SMEs and contribute to the growth of the AI-driven service robotics industry through international collaboration.

6. References

(1) Naoki Yokoyama, Alexander Clegg, Joanne Truong, Eric Undersander, Tsung-Yen Yang, Sergio Arnaud, Sehoon Ha, Dhruv Batra, Akshara Rai, “Adaptive Skill Coordination for Robotic Mobile Manipulation,” IEEE Robotics and Automation Letters (RA-L), 2023.

(2) Naoki Yokoyama, Sehoon Ha, Dhruv Batra, Jiuguang Wang, Bernadette Bucher, “VLFM: Vision-Language Frontier Maps for Zero-Shot Semantic Navigation,” IEEE International Conference on Robotics and Automation (ICRA), 2024. Best Conference Paper in Cognitive Robotics.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization